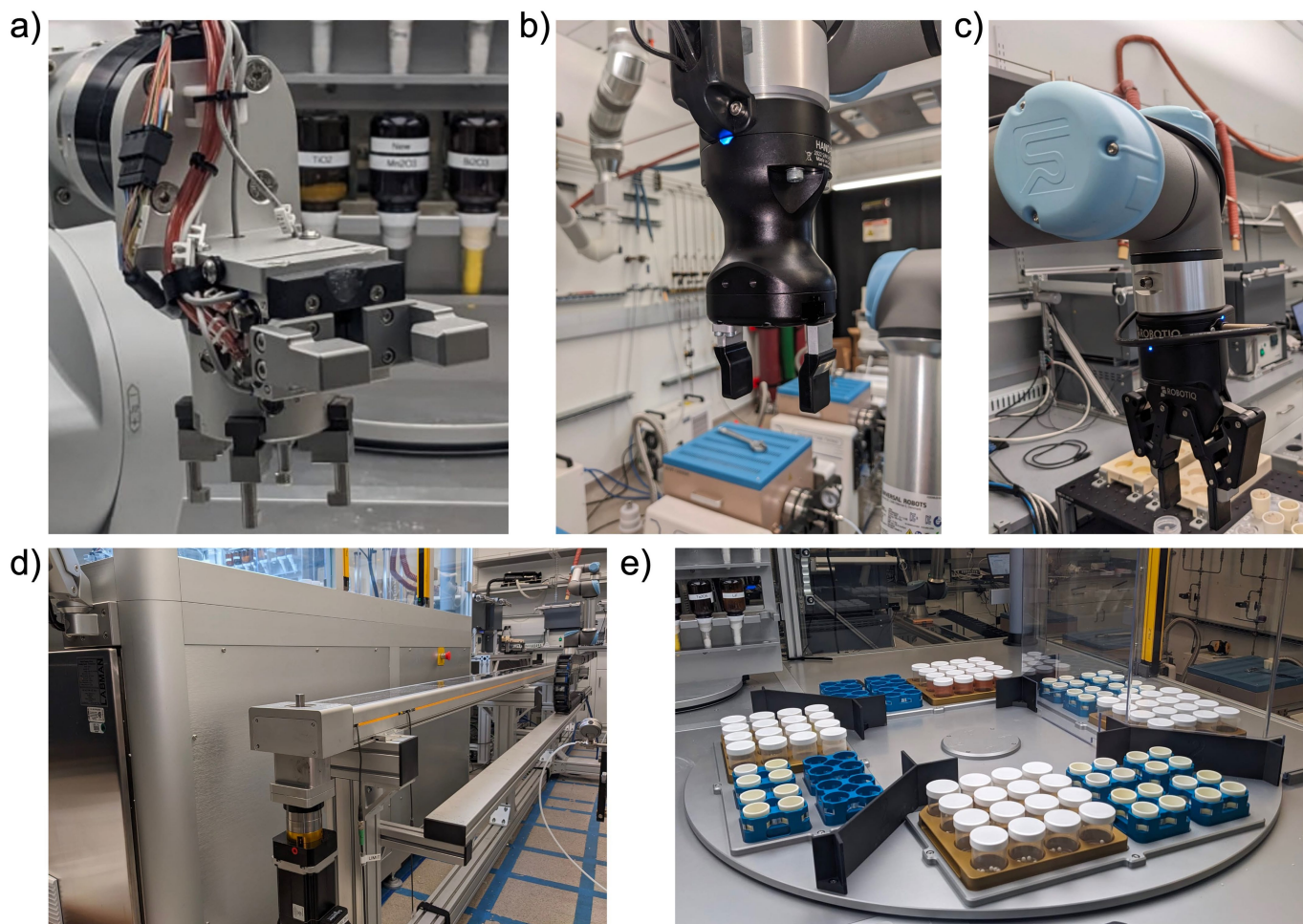
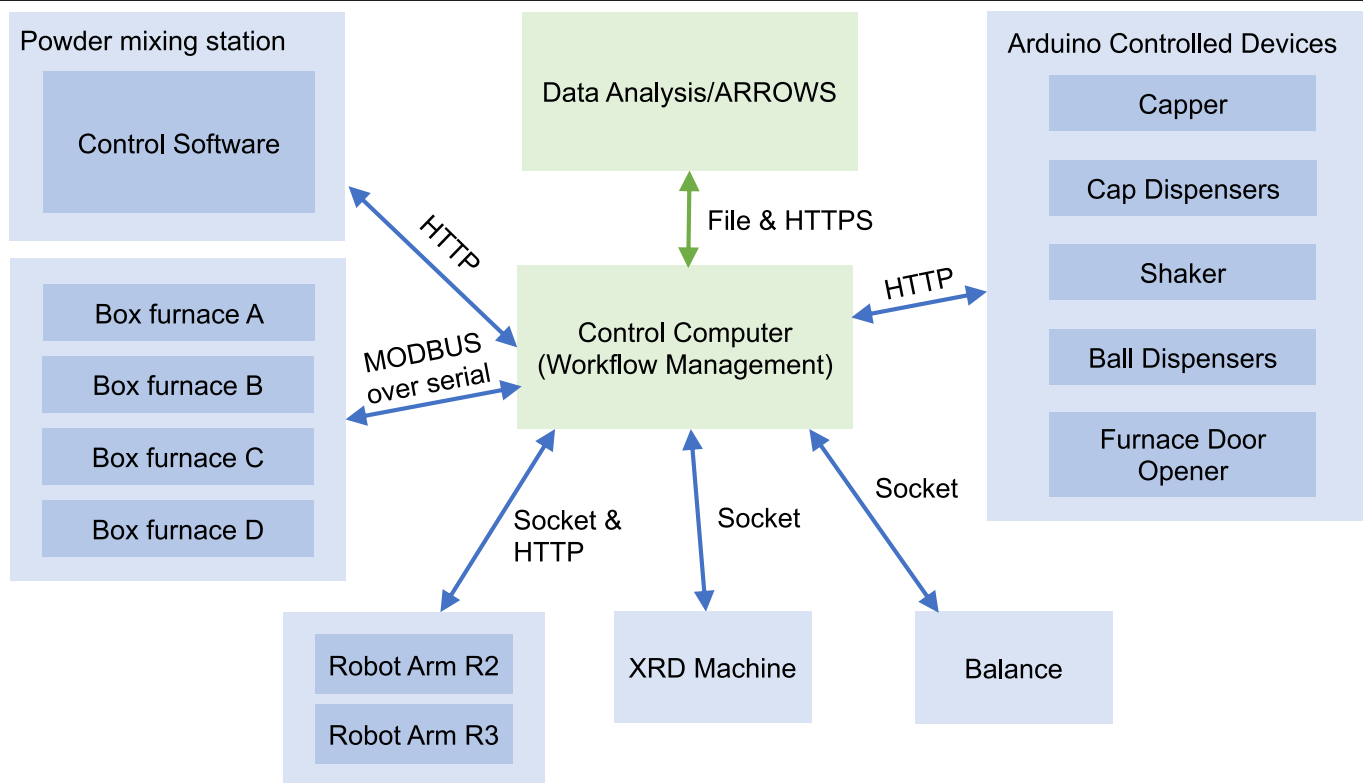




Extended Data Fig. 2 | Robotic installations for sample transfer in the A-Lab. Grippers on the UR5e robotic arms that are used for sample preparation (a), loading/unloading of crucible racks to/from the box furnaces (b) and sample retrieval and characterization (c). d, Linear rail used to increase the working

envelope of the robotic arm that loads/unloads crucible racks to/from the furnaces. e, Carousel used to organize and move samples in the sample preparation station.



Extended Data Fig. 3 | Communication protocols connecting each module in the A-Lab. A local area network (LAN) is built to connect all the pieces of the A-Lab with a control computer using an RS-485 interface (or DB25 for the box furnaces). Each module on the RS-485 interface has an Internet Protocol (IP)

assigned to enable communication with the computer. For enhanced cybersecurity, only the control computer has access to the internet, whereas the LAN is isolated from it.